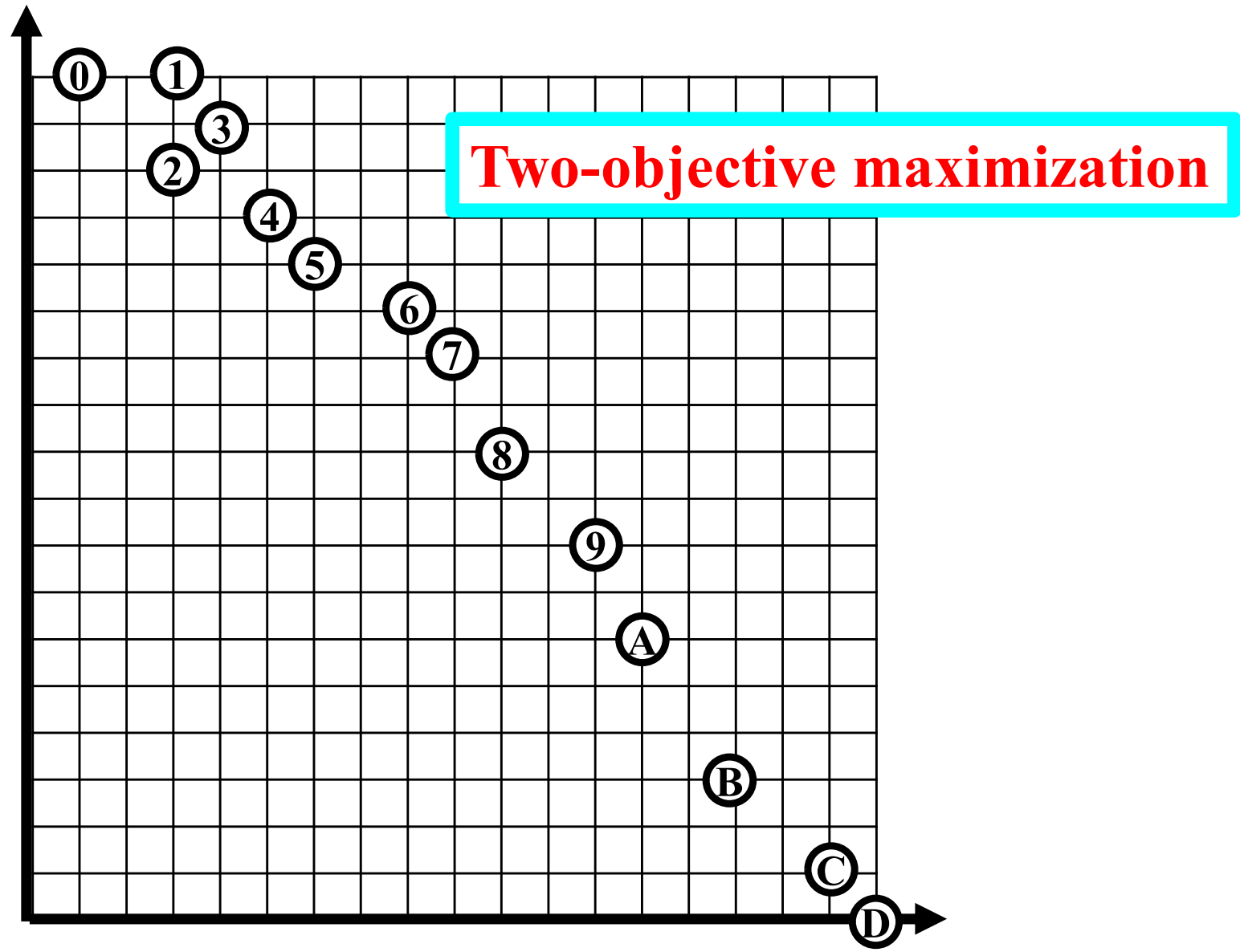
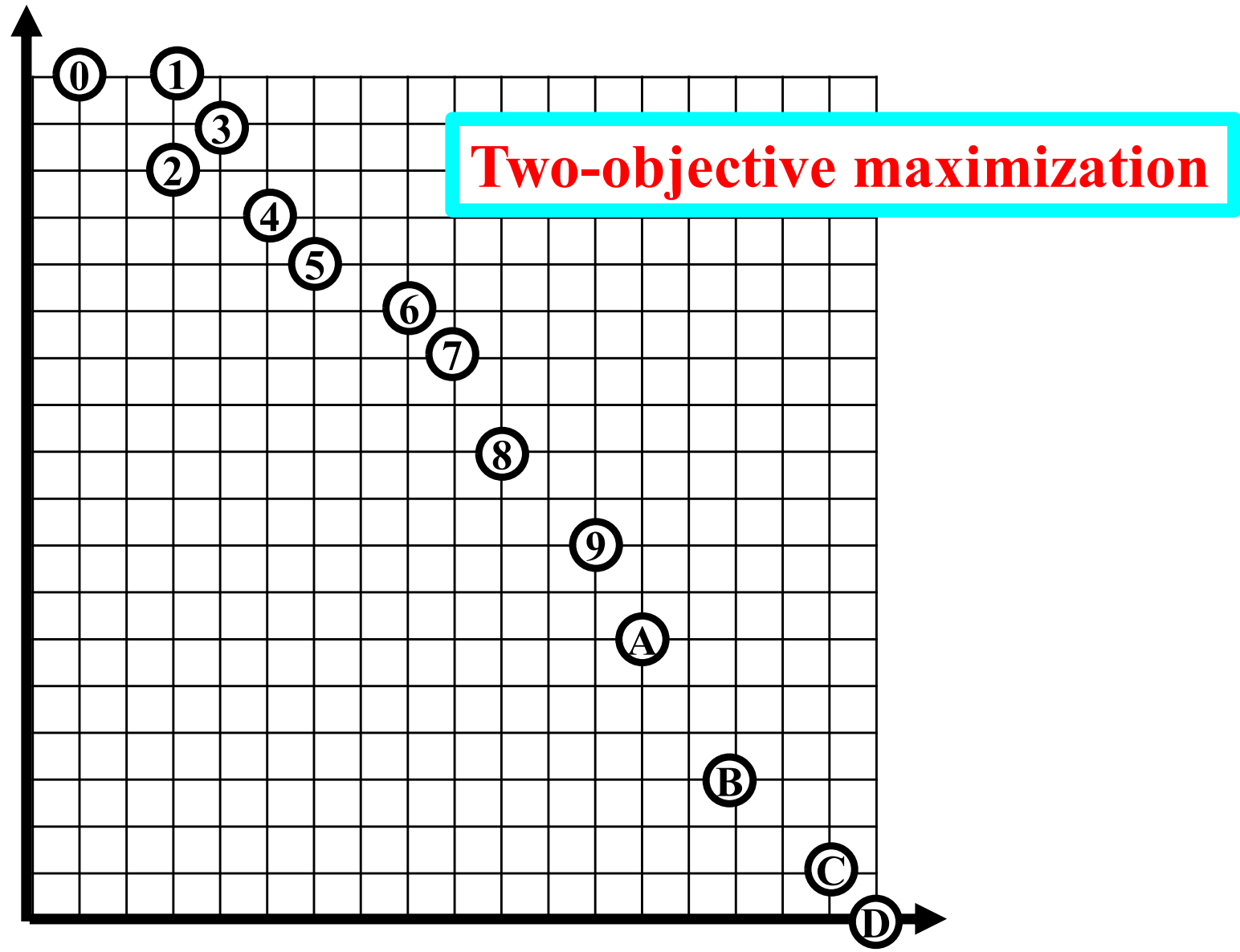


Lab Session. Task 1: We have 14 solutions at grid points. Using NSGA-II, choose 7 solutions for the next population.

Task 2: In a good EMO algorithm, which solutions should be selected (select 7 solutions for the next population using your own idea).



Task 3: Modify the generation update mechanism (i.e., the mechanism to select the next population) in order to choose more uniformly distributed solutions. That is, modify NSGA-II to address the issues shown by Task 1 and Task 2. (You need to explain your own idea)



Task 4: Even if we use the crowding distance recalculation, we cannot obtain a uniformly distributed solution set for three-objective problems by NSGA-II. This is because the crowding distance is calculated on each axis of the objective space. Please try to modify the generation update mechanism of NSGA-II for multi-objective problems with three or more objectives to find a uniformly distributed solution set (i.e., to modify the crowding distance or create a new mechanism).

